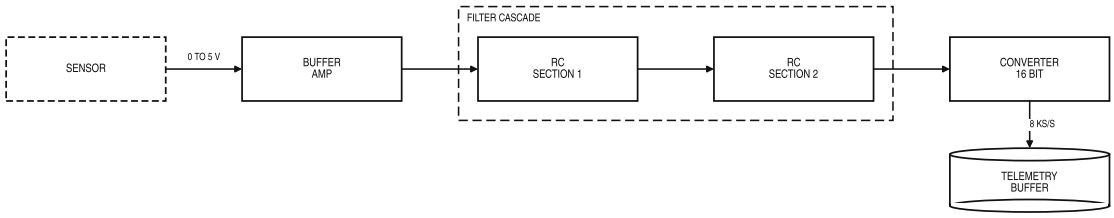
ANTI-ALIAS FILTER

an engineering section with a block diagram, math, a log-frequency plot and a parts table

4.2.1 FUNCTIONAL DESCRIPTION.

Each sensor channel passes through a cascade of first-order RC sections ahead of the converter. The cascade removes energy above the Nyquist frequency so that it cannot fold back into the measurement band.



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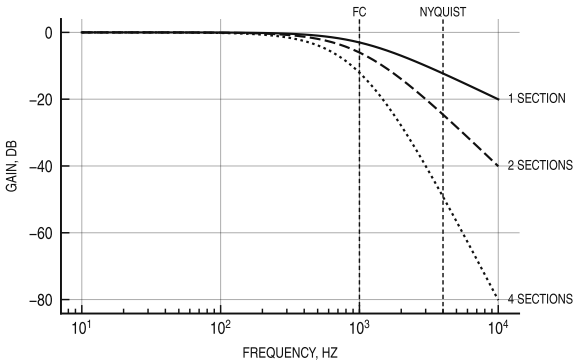
Figure 4.2-1. Signal Conditioning Channel, One of Eight

4.2.1.1 Transfer function.

One section passes a fraction of the input set by the ratio of frequency to the corner frequency f_c:

$$H(f) = \frac{1}{1 + jf/f_c}, \qquad f_c = \frac{1}{2\pi RC} \tag{4.2-1}$$

With $R = 1.6\ \text{k}\Omega$ and $C = 100\ \text{nF}$ the corner is $f_c \approx 1.0\ \text{kHz}$. Each section adds 6 dB per octave of rolloff above the corner (figure 4.2-2).



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Figure 4.2-2. Magnitude Response of One, Two and Four Sections

Table 4.2-1. Section Parts

Ref	Value	Tol	Note
R1, R2	1.6 k	1 %	metal film
C1, C2	100 nF	5 %	C0G
U1	--	--	rail-to-rail buffer

CAUTION

Do not substitute X7R capacitors. Their value falls with applied voltage and moves the corner.

4.2.2 VERIFICATION.

```
CHANNEL RESPONSE CHECK                                  TEST
SIG GEN ............... 1 KHZ, 1 V RMS
CHAN 1 OUTPUT ......... -3 DB +/- 0.5
SIG GEN ............... 4 KHZ
CHAN 1 OUTPUT ......... -24 DB OR LESS
REPEAT ................ CHAN 2 THRU 8
```